

Mitigating RAG-Induced Accuracy Collapse in a Zero-Resource Tulu Legal-Classification Pipeline

TL;DR

- Your failure is a textbook **context-memory / distraction failure**: irrelevant-but-fluent retrieved passages act as "hard distractors," and both of your sub-mechanisms (fact substitution, confabulation) are documented in the literature — the fix is NOT more prompt instructions (which the literature shows can even backfire), but a **pipeline-level relevance gate that removes bad passages before generation**.
- Given your exact constraints (local Llama-3 via Ollama, no fine-tuning, prompt/pipeline-only), the single best-supported first fix is a **confidence-gated + relevance-verification filter**: only inject a passage when retrieval similarity clears a tuned threshold AND a cheap second LLM/NLI call judges it relevant to the *person's situation*; otherwise fall back to your 80%-accurate no-RAG path. This is the mechanism validated by CRAG on 7B open-weight models.
- Two capability updates you may not know: **Ollama now exposes logprobs** (v0.12.11, Nov 2025) and **JSON-schema-constrained structured output** (`(format)` parameter on `(/api/generate)`), so confidence signals and field-separated output ARE available to you — but forcing rigid schemas on small models carries a measurable accuracy tax, so use structure carefully.

Key Findings

- 1. Your two sub-mechanisms are named, studied phenomena.** "Fact substitution" (adopting a wrong-but-real detail from an irrelevant passage) is the **distracting-passage / hard-negative** effect; "confabulation" grounded in neither source is **context-induced hallucination** driven by irrelevant context. Both worsen precisely when the query is low-resource/underspecified, because the model's parametric signal for the query is weak, so retrieved text dominates attention.
- 2. More prompting instructions will not save you — and can hurt.** You already give three explicit priority instructions. The literature is clear that instruction-only context-faithfulness has a ceiling and that reasoning/reiteration tricks can *increase* context reliance in a way that harms tasks needing the query to win. This is why your instruction is "given and not followed."
- 3. The empirically strongest, constraint-feasible fixes are pipeline-level filtering/gating, not decoding hacks.** CRAG (a lightweight relevance evaluator + fallback) is validated on 7B open models. Confidence-gated retrieval is validated with the finding that injecting retrieval on already-answerable queries *degrades* accuracy — exactly your 80%→25% pattern.
- 4. Ollama exposes more than you're currently using.** Structured JSON output, per-token logprobs, and multiple sampled completions for self-consistency are all available.

Details

A. What is actually going wrong (diagnosis)

The RAG "power of noise" literature establishes that **random irrelevant passages are relatively harmless, but *distracting* passages — semantically related, fluent, containing specific but wrong details — actively degrade accuracy**. Amiraz et al. ("The Distracting Effect," arXiv 2505.06914, ACL 2025 Long Papers, pp. 18228–18258) formalize this and quantify it precisely: hard distracting passages make "the accuracy drop from 6 to 11 accuracy points, depending on the LLM" (measured on Natural Questions with Llama-3.1-8B), and they report that over 60% of top-10 dense-retriever results contain at least one hard distractor. This matches your setup exactly: retrieval has above-chance topical accuracy, so it returns on-topic legal cases that nonetheless contain the wrong specific facts (mother vs. father), which the model then substitutes. (arXiv)

(arXiv)

Your **confabulation** case (content in neither query nor the 24,920-char passage) is the classic context-triggered hallucination: the mere presence of retrieved legal text shifts the model into "narrate a plausible legal case" mode, and with a weak low-resource query signal it invents the most statistically typical legal scenario (death/inheritance). NoMIRACL (Thakur et al., EMNLP 2024 Findings, pp. 12508–12526) documents that small open models hallucinate massively when all passages are non-relevant: "Models such as LLAMA-2 and Orca-2 achieve over 88% hallucination rate on the non-relevant subset." Importantly, the *same* paper shows this is strongly model-dependent — Llama-3-8B-Instruct averages only 26.8% and Mixtral-8x7B 17.4%, versus GPT-4 at 35.5% — so the specific Llama-3 you run should be less pathological than Llama-2, and the residual failures you see are likely dominated by the *distractor* mechanism (on-topic wrong passages) rather than pure non-relevance. This is the single most important external validation that your problem is expected behavior, not a bug in your code. (ACL Anthology) (Hugging Face)

The mechanism is worse for the low-resource query condition because context-memory conflict is **task- and signal-dependent**: when the model has little parametric grounding for the query (a zero-resource Tulu situation), the retrieved context wins by default (consistent with "Task Matters," arXiv 2506.06485, which finds context-reliance strategies help context-only tasks but harm tasks that need parametric/query knowledge to dominate).

B. The five candidate techniques, rated by evidence and feasibility

(a) Relevance-verification step — STRONGEST EVIDENCE, FEASIBLE. RECOMMENDED.

- *What it is*: Before using a passage, an explicit judgment (separate LLM call or a small NLI model) of whether it is relevant; discard/down-weight if not.
- *Validation*: This is the core of **CRAG (Corrective RAG; Yan et al., arXiv 2401.15884, ICLR 2024)**. Crucially for you, CRAG was tested on **7B open models (LLaMA2-hf-7B and SelfRAG-LLaMA2-7B), not just GPT-4**, using a lightweight 0.77B T5-large retrieval evaluator. Per its Table 5, on SelfRAG-LLaMA2-7B, CRAG beat vanilla RAG by "7.0%

accuracy on PopQA, 14.9% FactScore on Biography, 36.6% accuracy on PubHealth, and 15.4% accuracy on Arc-Challenge"; on plain LLaMA2-hf-7B by 4.4% (PopQA), 2.8% (Biography), and 10.3% (Arc-Challenge). An open reproduction (arXiv 2603.16169) confirmed transfer to a smaller generator, Phi-3-mini (~3.8B): "achieves 54.4% on PopQA, closely matching the original system's 54.9%," and 85.2% vs 84.8% vanilla RAG on ARC-Challenge. **Yoran et al. (ICLR 2024, arXiv 2310.01558)** show an NLI filter that drops passages not entailing the answer "is effective in preventing performance reduction" — though it "is too strict" and discards some relevant passages. (arXiv + 4)

- **Complexity:** Multi-call pipeline (one extra LLM call per passage, or one batched call), OR a small offboard NLI/reranker model. No fine-tuning, no logits needed.

(b) Self-consistency / no-RAG vs RAG cross-check — GOOD EVIDENCE, FEASIBLE. RECOMMENDED as complement.

- **What it is:** Generate the no-RAG answer (your 80% baseline) and the RAG answer; when they disagree, don't default to RAG — flag, or prefer the no-RAG answer / abstain.
- **Validation:** Self-consistency (Wang et al. 2022) and disagreement-based selection are well established. "Disentangling sources" (generate the parametric answer first, then present it alongside the evidence) is an evaluated context-memory mitigation, tested on Qwen2.5-7B-Instruct and Llama-3.1-8B-Instruct (arXiv 2601.11886). Given your no-RAG path is 80% and RAG is 25%, disagreement is a strong wrong-answer signal.
- **Complexity:** Multi-call (2+ calls); trivial to implement. Ollama supports multiple sampled completions (repeat calls with temperature > 0).

(c) Attributed / citation-forcing prompting — MODERATE EVIDENCE, FEASIBLE.

- **What it is:** Require the model to quote the exact span of the passage supporting each claim; unsupported claims become visible/blockable.
- **Validation:** Zhou et al. ("Context-faithful Prompting," EMNLP 2023 Findings, arXiv 2303.11315) — attributed and opinion-based prompts significantly improve faithfulness, tested on GPT-3.5 and Llama-2-7B-chat. Citation-enforced medical-RAG work ("StrictCitations," MDPI Applied Sciences 16:3013) found citation enforcement exposes and discourages unsupported internal knowledge, but also that Llama-3.1 and Gemma-2 sometimes still override context ("knowledge override") — so it's a detector, not a cure. Note the twist: your problem is over-reliance on context, whereas citation-forcing is usually deployed against under-reliance; here its value is that a confabulation cannot cite a real span, so it becomes catchable in a post-processing check.
- **Complexity:** Single prompt rewrite + a post-processing check that the quoted span literally appears in the passage.

(d) Confidence-gated retrieval — STRONG EVIDENCE, FEASIBLE. RECOMMENDED as the gate.

- *What it is:* Only inject retrieval when retrieval similarity/confidence clears a threshold; below it, fall back to no-RAG.
- *Validation:* CONQORD (arXiv 2404.17287) reports that adding retrieval to high-confidence (already-answerable) responses "may lead to unexpected performance degradation ... due to ... additional information noise" — your exact pathology. A zero-shot small-LLM confidence study (arXiv 2605.02241) found strong-hit-only retrieval (threshold $\tau \approx 0.70$) improved its confidence signal on Llama, while including marginal hits ($\tau \approx 0.60$) "degrade[d] it below the bare baseline" — direct evidence that a well-chosen similarity threshold matters and that marginal passages are net-harmful.
- *Complexity:* Trivial — a threshold on the score you already compute. This is the cheapest single change.

(e) Structured output separating query-facts from passage-facts — PLAUSIBLE, FEASIBLE, USE WITH CARE.

- *What it is:* Force distinct fields ("what the person's situation says" vs "what the passage says" vs "final label") instead of blended free-text reasoning.
- *Validation:* Indirect. Ollama supports JSON-schema-constrained output. BUT the "Constraint Tax" work (arXiv 2605.26128) shows forcing hard schemas on small models can *lower* answer accuracy substantially — across 15,000 generations on Qwen2.5-0.5B/1.5B and SmoLLM2-1.7B, hard schema decoding raised validity from 61.5% to 100% but dropped answer accuracy from 19.7% to 11.0% (−8.7 points), with a tool-call analogue falling from 91.5% to 48.0% executable accuracy despite both being 100% schema-valid. Best practice from that literature: "**reason free, constrain late**" — let the model reason, then extract into structure in a second call, rather than constraining the reasoning itself. [arxiv](#) [arxiv](#)
- *Complexity:* Single-to-moderate. Feasible but not your highest-leverage first move.

C. Ollama capability audit (directly answering your constraint questions)

- **Structured/JSON-schema output: YES.** The `format` parameter accepts a full JSON schema on both `/api/chat` and `/api/generate`; this is grammar-constrained decoding under the hood (guaranteed schema-valid). You can enforce separate fields and an enumerated label set.
- **Logprobs / token probabilities: YES, as of Ollama v0.12.11 (released Nov 2025).** Both the native API (`logprobs: true`, `top_logprobs: N` on `/api/generate`) and the OpenAI-compatible endpoint now return per-token logprobs. Confidence signals ARE therefore available to you now — a change from the long-standing gap (GitHub issue #2415). Caveat: it returns only the top-N candidates, so any confidence is normalized over the visible set, and the aggregation formula you choose matters more than N. This partially relaxes your "no logit access" constraint, but confidence-gating on *retrieval similarity* (which you already compute) is simpler and better-validated for your case than logprob-based gating.

- **Multiple sampled completions for self-consistency: YES** — call repeatedly with temperature > 0 (or higher `top_p`).
- **Grammar-constrained generation: YES** (via `format`)/underlying GBNF-style constraint).
- **Fine-tuning / weight access: not needed** for any recommended technique.

D. Transfer to your exact setting (low-resource + legal + small local model)

- *Low-resource/cross-lingual*: NoMIRACL is explicitly multilingual (18 typologically diverse languages) and shows the abstention/relevance problem is language-general, with small open models the worst offenders. This is strong reason to expect a relevance gate to help you specifically. Broader multilingual-RAG surveys confirm retrieval and abstention are harder in low-resource languages.
- *Legal domain*: Legal RAG hallucination is heavily documented. Magesh et al. (2025, *Journal of Empirical Legal Studies* 22:216–242, testing May-2024 tools) found hallucination rates of 17% for Lexis+ AI, 33% for Westlaw AI-Assisted Research, and 43% for GPT-4 — i.e., even purpose-built commercial legal RAG hallucinates at 17–33%. Dahl et al. ("Large Legal Fictions," arXiv 2401.01301) report that "legal hallucinations are alarmingly prevalent, occurring between 69% of the time with ChatGPT 3.5 and 88% with Llama 2, when these models are asked specific, verifiable questions about random federal court cases." Legal-RAG work consistently concludes relevance filtering plus human-verifiable attribution are necessary and that hallucination cannot be fully eliminated. (AI Law Librarians) (Stanford Law School)
- *Small local models (Llama-3-8B class)*: CRAG (7B), Open-RAG (arXiv 2410.01782, open models), and the zero-shot small-LLM confidence work (arXiv 2605.02241) all validate filtering/gating on your model class — the key point being that these are NOT GPT-4-only results.

Recommendations

Stage 1 — Ship the cheapest gate first (a few hours; tests the core hypothesis). Implement **confidence-gated retrieval (d)**: use your existing retrieval similarity; if the top passage's score is below a threshold τ , **skip RAG entirely and use the no-RAG path** (already 80%). Sweep τ on your $n=20$ set. This alone should recover most of the 80%→25% collapse if, as suspected, many queries retrieve marginal/distracting passages. **Benchmark that changes the plan**: if accuracy with the gate approaches the 80% baseline, the problem was noise injection and you're largely done; if it stays low even when only high-similarity passages pass, the problem is genuine distraction from *on-topic* passages and you must proceed to Stage 2.

Stage 2 — Add a relevance-verification call (the highest-evidence fix). Add a CRAG-style **binary relevance judgment (a)** as a separate Ollama call: "Does this passage describe the SAME situation as the person's query? Answer RELEVANT or NOT_RELEVANT." Discard NOT_RELEVANT passages; if none survive, fall back to no-RAG. Keep this call simple (single-

token answer), not a rigid schema. This is the intervention with the cleanest 7B-model empirical support (CRAG). **Threshold to change plan:** if the relevance judge is itself unreliable on Tulu queries (check its precision/recall against your gold relevance labels), switch the judge to operate on an English translation of the query, or replace it with an NLI/cross-encoder reranker run offboard.

Stage 3 — Add a disagreement guard (cheap insurance). Run **no-RAG vs RAG self-consistency (b)**: if the two answers disagree, prefer the no-RAG label or abstain. Given your baseline asymmetry (80% vs 25%), disagreement is a high-value wrong-answer flag, and it is nearly free to add on top of Stages 1-2.

Do first, do NOT do first:

- **DO** pick ONE technique (Stage 1, then Stage 2) and evaluate it properly. $n=20$ is too small for multi-arm comparisons — report exact-match with confidence intervals or per-item deltas and treat results as directional.
- **DON'T** rewrite the prompt again hoping for compliance — you've already shown the instruction is ignored.
- **DON'T** lead with rigid JSON-schema reasoning (e); the constraint tax on small models can cost accuracy. If you want field separation, "reason free, then extract in a second constrained call."
- **DON'T** rely on logprob-based confidence as your primary gate; retrieval-similarity gating is simpler and better-validated for your case.

Caveats

- **$n=20$ is a severe limitation.** Frame every recommendation in the paper as a directional pilot; do not report statistical significance. Consider expanding the test set even modestly.
- **CRAG/Yoran/CONQORD gains are on QA benchmarks (PopQA, Arc, TriviaQA, etc.), not Tulu legal classification.** Transfer is plausible and well-motivated but unproven for your exact task — that gap is precisely the contribution your NLLP paper can make.
- **The relevance judge is itself an LLM call on a low-resource language**, so it may inherit the same Tulu weakness; validate it independently and consider judging on translated text.
- **Yoran's NLI filter was validated on Llama-2-13B/70B, not 8B** — flagged so you don't overclaim its transfer to your model size.
- **Ollama logprobs are recent (v0.12.11) and top-N truncated**; verify your installed version and treat reconstructed probabilities as approximate.
- **Structured-output constraint-tax figures come from sub-3B models**; the effect on Llama-3-8B may be smaller, but the "reason free, constrain late" principle still applies.
- Legal-domain hallucination percentages and the distraction-effect magnitudes come from the primary studies cited (Magesh et al. 2025; Dahl et al. arXiv 2401.01301; Amiraz et al.

arXiv 2505.06914); verify the exact figures against those primaries before citing in a camera-ready paper.